

Building a Neural Network in TensorFlow

While you can compute the output of a network one layer at a time, TensorFlow provides a much simpler and more powerful way to build, train, and run models using the **Sequential API**.

The Sequential Model: A Simpler Approach

Instead of manually passing the output of one layer to the input of the next, the `Sequential` model allows you to define a "stack" or sequence of layers. TensorFlow then handles the work of stringing them together.

- **Old Method (Manual Forward Prop):**

```
1 a1 = Layer1(x)
2 a2 = Layer2(a1)
3 a3 = Layer3(a2)
```

- **New Method (Using `Sequential`):**

You define all the layers at once, and TensorFlow connects them for you.

A Common Coding Convention 🏠

The most common way to build a model is to define the layers directly *inside* the `Sequential` function.

- **Example (Digit Classification):**

```
1 import tensorflow as tf
2 from tensorflow.keras.layers import Dense
3 from tensorflow.keras.models import Sequential
4
5 model = Sequential([
6     Dense(units=25, activation='sigmoid'), # Layer 1
7     Dense(units=15, activation='sigmoid'), # Layer 2
8     Dense(units=1, activation='sigmoid')  # Layer 3 (Output)
9 ])
```

- This single `model` object now contains the entire three-layer network architecture.
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How to Use the Sequential Model

Once you have a `Sequential` model, training it and making predictions becomes simple.

- **1. Training the Model (Preview):**

- `model.compile(...)`: A required setup step to configure the model for training (details will be covered next week).
- `model.fit(X, Y)`: A single command to **train** the network. You provide the training data (`X`) and the target labels (`Y`), and TensorFlow handles the entire gradient descent process.

- **2. Making Predictions (Inference):**

- `model.predict(X_new)`: This is the new way to perform forward propagation. You pass in a new input (`X_new`), and the model automatically runs it through all the layers and returns the final prediction.
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Understanding "Under the Hood"

- While frameworks like TensorFlow allow you to build complex neural networks in just a few lines of code, it is critical to understand *what* those lines are actually doing.
- Most engineers use libraries, but the best engineers understand the underlying mechanics. To build this understanding, the next step is to learn how to implement forward propagation from scratch.